MIT SCHOOL OF COMPUTING
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MIT ART, DESIGN & TECHNOLOGY UNIVERSITY
PUNE, INDIA
A Leap Towards World Class Education

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Subject: IDS Roll No.: 43 Exp. No.: 2
Name of the Experiment:
Performed on:
Submitted on:

Marks	Teacher's Signature with date
10	

1 Sequential Organiszation & Characteristics of linear data Structures.

→ It is a method of Storing data elements in a Continous location where elements in a continous memory location elements are arranged and after another in a fixed order.

Characteristics:

- elements are Stored in contingous memory locations.
- each element has a unique index or position
- Data is arranged in a linear or position
- Size is usually fixed.

Incomplete for Theory ☐ Diagrams ☐ Observation Table ☐
Calculations ☐ Graphs ☐ Results ☐ Conclusion ☐
Understanding Through Q/A ☐ Late Submission ☐ Neatness ☐

Teacher's Signature

2. Array as an abstract Data type (ADT)

ADT is Collection of elements of the Same data type stored in Contingous memory locations. & accessed using an index

o Components of ADT :

→ Elements : Data items of the Same type.
→ Index : Position used to access element
→ Size : total number of elements.

o Basic Operations on array :

→ Transversal : Visiting each element
→ Insertion : Adding a new element
→ Deletion : remaining an element
→ Searching : finding an element.

3. Insertion, deletion & Searching in a 1D array.

Consider an array:

A = [10, 20, 30, 40]

insertion : insert 15 an Position 2 :

Shift elements right → A = [10, 20, 25, 30, 40]

- Deletion : Delete element at Position 3:

  Shift elements left → A = [10, 20, 25, 40]

- Searching : To search 25

  Compare Sequentially until found.

  25 is found in Index 2.

4 Advantages & limitations of arrays

Advantages:

- fast direct access using index (O(1))
- Simple & easy to implement
- efficiently memory usage for fixed size data.
- good for storing homogenous data.

limitations:

- fixed size
- Insertion & deletion are time - Constant
- Wastage of memory if size not useful

efficiently comparision:

- Insertion : O(n) due to shifting
- Deletion : O(n) due to shifting
- Searching: O(n) for linear search.
  O(log n) for binary search (sorted array).

5] 2-Dimentional array for student academic data:

2D array: Rows represent students
Columns represent marks / subject

Ex:-

Student	Math	Science	English
S1	85	90	88
S2	78	82	80

Accessing data:

- Marks [0] [1] -> Science Marks
- traversal uses nested loops

efficient manipulation.

- Direct indexing allows fast access.
- Nested loop helps in processing total & Averages.

Extension to n dimentional arrays:

- image Processing (3D arrays for RGB)
- Scientific Simulations
- ML & data analysis
- game development